

Chapter 1: Preliminaries

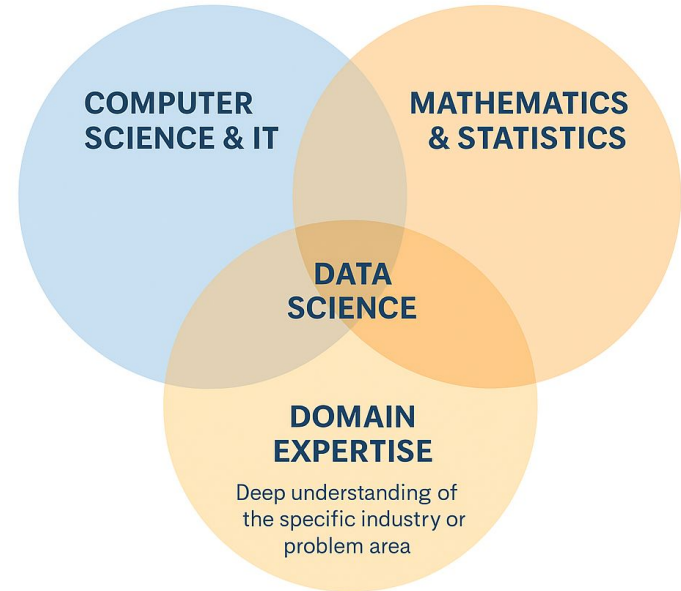


Outlines

1. Introduction
2. Linear Algebra Review
3. Probability
4. Statistics

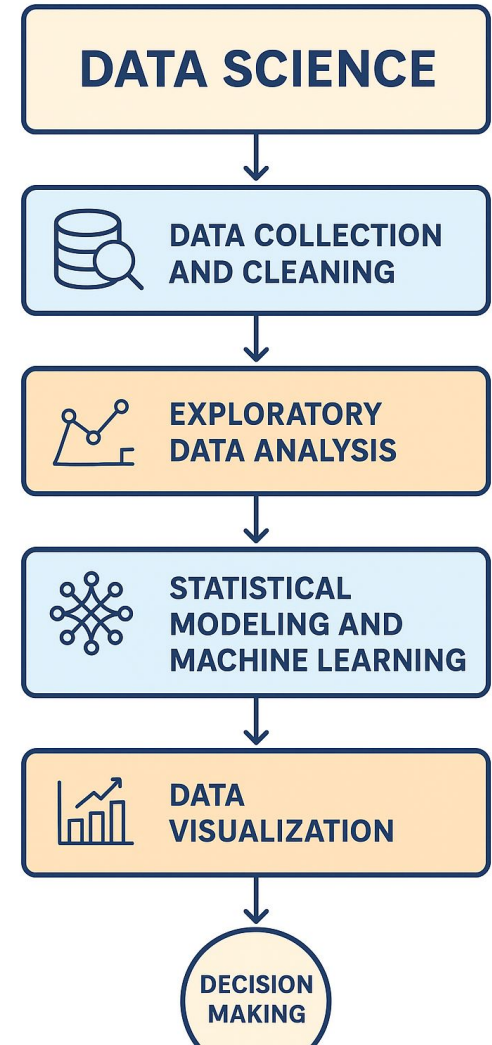
Introduction

- Data science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data.
- Data Science industry is based on three main sources of knowledge:
 - Mathematics & Statistics
 - Computer Science & Information technology
 - Domain Knowledge



Introduction

- A data scientist is someone who extracts insights from messy data.



Linear Algebra Review

Linear algebra is the branch of mathematics that deals with vector spaces.

Vectors

An ordered sequence $(a_1, a_2, \dots, a_n)$ of numbers is called an ordered n-tuple, a vector.

Vector Addition

$$\vec{u} = (u_1, u_2, \dots, u_n)$$

$$\vec{v} = (v_1, v_2, \dots, v_n)$$

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

Vectors

Scalar Multiplication

$$\vec{v} = (v_1, v_2, \dots, v_n)$$

$$k\vec{v} = (kv_1, kv_2, \dots, kv_n)$$

Matrices

- A matrix is a two-dimensional collection of numbers.
- Represent matrices as lists of lists, with each inner list having the same size and representing a row of the matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$A = \begin{bmatrix} [1, 2, 3], \\ [4, 5, 6] \end{bmatrix}$ *# A has 2 rows and 3 columns*

$B = \begin{bmatrix} [1, 2], \\ [3, 4], \\ [5, 6] \end{bmatrix}$ *# B has 3 rows and 2 columns*

Matrices

Matrix Addition

- If A and B are matrices of the same size, their sum $A + B$ is the matrix formed by adding corresponding entries.
- If $A = [a_{ij}]$ and $B = [b_{ij}]$, this takes the form $A + B = [a_{ij} + b_{ij}]$

Matrices

Scalar Multiplication

- If A is any matrix and k is any number, the scalar multiple kA is the matrix obtained from A by multiplying each entry of A by k .
- If $A = [a_{ij}]$, this is $kA = [ka_{ij}]$

Matrices

Matrix Multiplication

Let A be an $\mathbf{m} \times \mathbf{n}$ matrix, let B be an $\mathbf{n} \times \mathbf{k}$ matrix, $C=AB$

$$C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1k} \\ c_{21} & c_{22} & \dots & c_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mk} \end{bmatrix}$$

$$c_{ij} = \sum_{r=1}^n a_{ir} \cdot b_{rj}$$

Matrices

Matrix Multiplication

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & -1 & -2 \end{bmatrix}, B = \begin{bmatrix} 1 & -2 \\ -2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 2 & -9 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -2 \\ -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 2 & -1 & -2 \end{bmatrix} = \begin{bmatrix} -3 & 3 & 7 \\ 0 & -3 & -8 \\ 5 & -1 & -1 \end{bmatrix}$$

$$c_{ij} = \sum_{r=1}^n a_{ir} \cdot b_{rj}$$

Transpose of a Matrix

Definition

- If A is an $m \times n$ matrix, the transpose of A , written A^T , is the $n \times m$ matrix whose rows are just the columns of A in the same order.

Exercises

Write python programs that fulfill the following requirements:

- (a) Sum two matrices, multiply a matrix by a number, multiply two matrices.
- (b) Calculate a transpose of a matrix.

Note: Each request is made in two ways:

1. code directly according to definitions, theorems, constructions,...
2. uses python libraries (e.g., numpy)

Probability Review

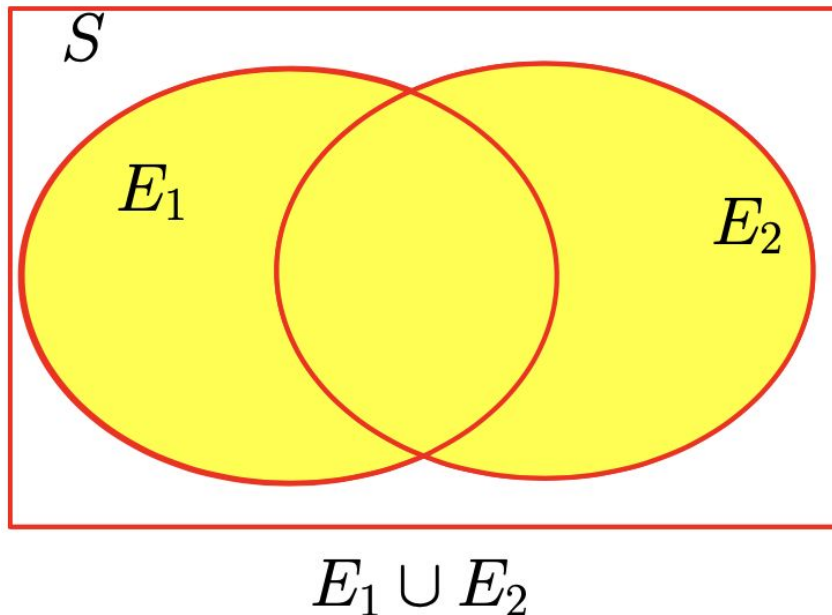
Sample Spaces and Events

Definition

- An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a **random experiment**.
- The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as **S**.
 - A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes.
 - A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.
- An **event**, denoted by **E**, is a subset of the sample space of a random experiment.

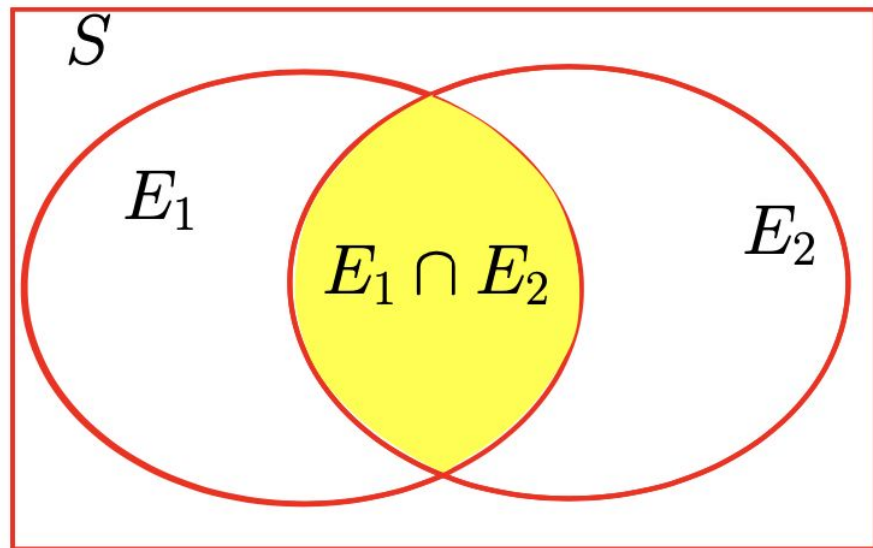
Sample Spaces and Events

- The **union** of two events is the event that consists of all outcomes that are contained in either of the two events.
- We denote the union as $E_1 \cup E_2$ or $E_1 + E_2$.



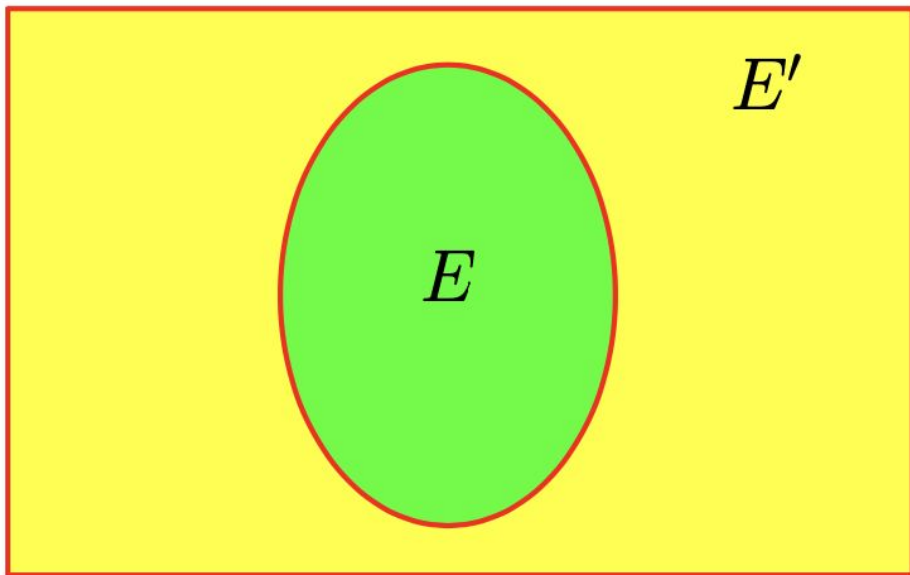
Sample Spaces and Events

- The **intersection** of two events is the event that consists of all outcomes that are contained in both of the two events.
- We denote the intersection as $E_1 \cap E_2$ or E_1E_2 .



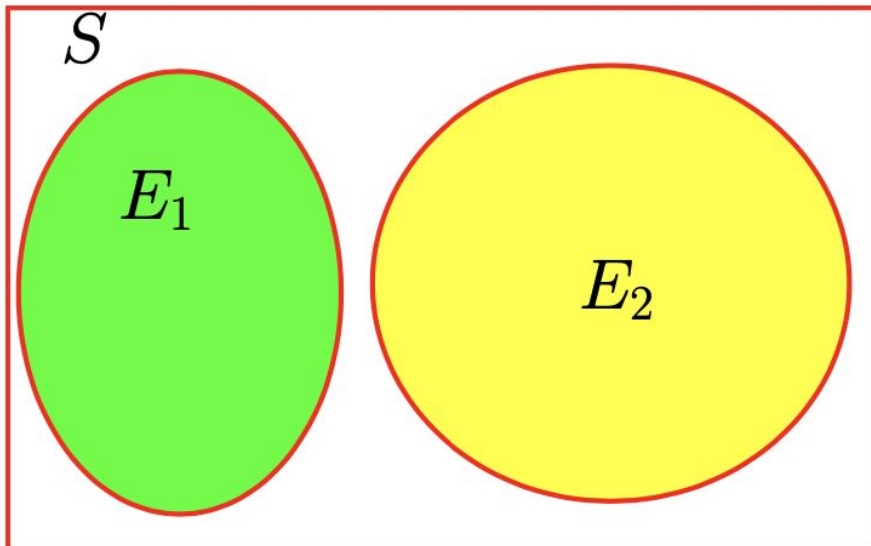
Sample Spaces and Events

- The **complement** of an event in a sample space is the set of outcomes in the sample space that are not in the event.
- We denote the complement of the event E as E' .



Sample Spaces and Events

- Two events, denoted as E_1 and E_2 , such that $E_1 \cap E_2 = \emptyset$.
are said to be **mutually exclusive**.



Addition Rules

- Any two events E_1, E_2

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

- Any three events E_1, E_2, E_3

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$$

Addition Rules

- If A and B are mutually exclusive events,
$$P(E1 \cup E2) = P(E1) + P(E2)$$
- A collection of events, $E1, E2, \dots, Ek$, is said to be mutually exclusive if for all pairs,
$$Ei \cap Ej = \emptyset$$
- For a collection of mutually exclusive events,
$$P(E1 \cup E2 \cup \dots \cup Ek) = P(E1) + P(E2) + \dots + P(Ek)$$

Conditional Probability

- The probability of an event B under the knowledge that the outcome will be in event A is denoted as $P(B|A)$ and this is called the **conditional probability** of B given A, and is calculated by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Multiplication and Total Probability Rules

- Multiplication rule for any two events A and B:

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

- Total probability rule for any two events A and B:

$$P(B) = P(B \cap A) + P(B \cap A') = P(B|A)P(A) + P(B|A')P(A')$$

- Assume $E_1, E_2, \dots, E_k$ are k mutually exclusive and exhaustive sets. Then

$$\begin{aligned} P(B) &= P(B \cap E_1) + P(B \cap E_2) + \dots + P(B \cap E_k) \\ &= P(B|E_1)P(E_1) + P(B|E_2)P(E_2) + \dots + P(B|E_k)P(E_k) \end{aligned}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

Independence

- Two events are independent if any one of the following equivalent statements is true:

$$P(A|B)=P(A)$$

$$P(B|A)=P(B)$$

$$P(A \cap B) = P(A)P(B)$$

Bayes' Theorem

If A, B are any two events:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \text{ for } P(B) > 0$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Bayes' Theorem

If A, B are any two events:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \text{ for } P(B) > 0$$

Random Variables

Definition

- A **random variable** is a variable whose possible values have an associated **probability distribution**.
- A **discrete random variable** is a random variable with a finite (or countably infinite) range.
- A **continuous random variable** is a random variable with an interval (either finite or infinite) of real numbers for its range.

Probability Distributions and Probability Density Functions

For a continuous random variable X , a probability density function is a function such that

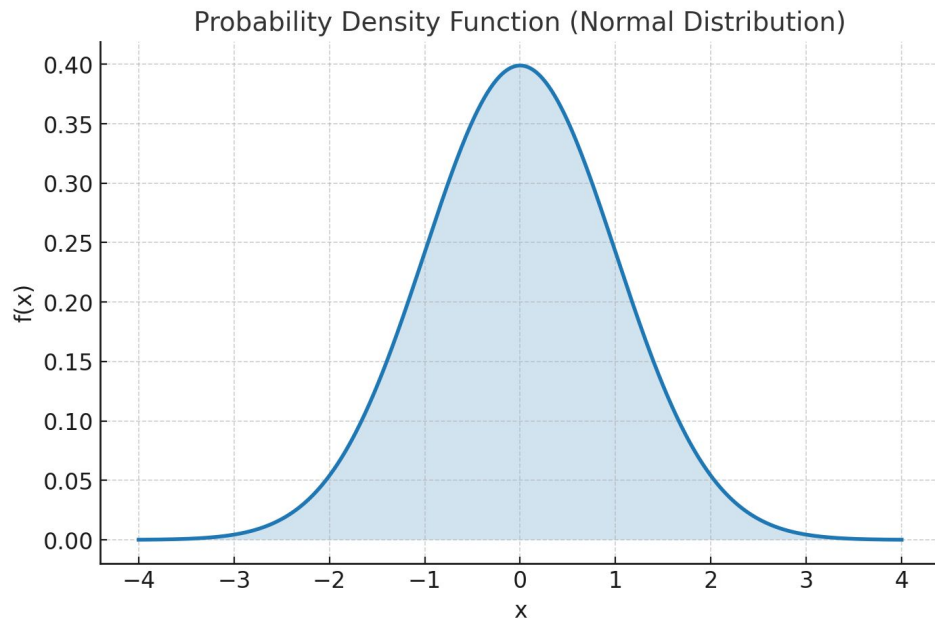
- Probability can't be negative.
- The total probability over all possible values is 1.

$$(1) f(x) \geq 0$$

$$(2) \int_{-\infty}^{\infty} f(x)dx = 1$$

The curve shows how probability is distributed across values of x .

The total area under the curve = 1.



Mean and Variance

X is a continuous random variable with probability density function $f(x)$.

* The **mean** or **expected value** of X, denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

* The **variance** of X, denoted as $V(X)$ or σ^2 , is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$$

* The **standard deviation** of X is

$$\sigma = \sqrt{\sigma^2}.$$

Continuous Uniform Distribution

- A continuous random variable X with probability density function

$$f(x) = 1/(b-a), a \leq x \leq b$$

is a **continuous uniform random variable**.

- If X is a continuous uniform random variable over $a \leq x \leq b$,

$$\mu = E(X) = \frac{(a+b)}{2} \text{ and } \sigma^2 = V(X) = \frac{(b-a)^2}{12}$$

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

Continuous Uniform Distribution

- The cumulative distribution function of a continuous uniform random variable is the probability that a random variable is less than or equal to a certain value.

$$F(x) = \begin{cases} 0 & x < a \\ (x - a)/(b - a) & a \leq x < b \\ 1 & b \leq x \end{cases}$$

Normal Distribution

Normal Distribution (also called the Gaussian Distribution)

- A continuous random variable X follows a normal distribution with mean μ and standard deviation σ if its probability density function (PDF) is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- The Normal Distribution is the bell curve-shaped distribution and is completely determined by two parameters:
 - μ : indicates where the bell is centered
 - σ : how “wide” the bell is

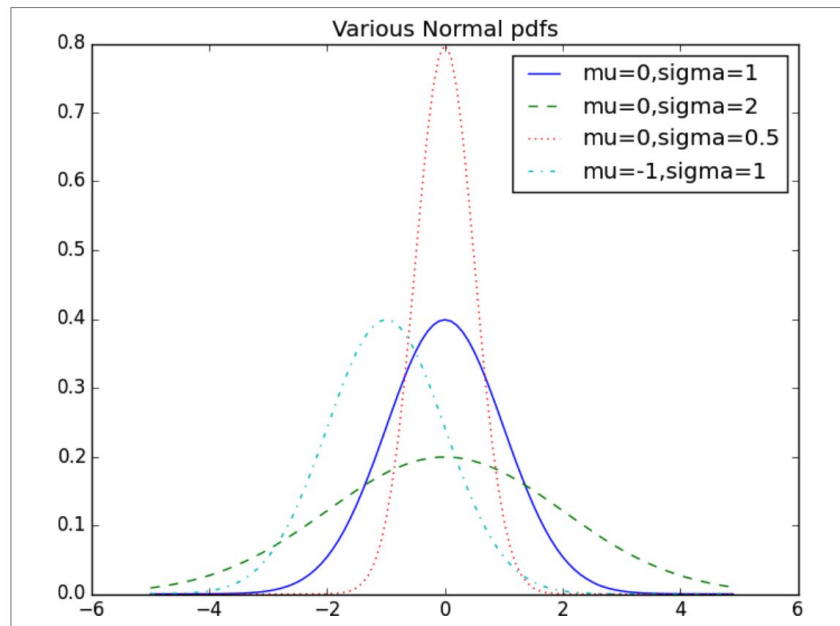


Figure 6-2. Various normal pdfs

Normal Distribution

Standard Normal Distribution

- A normal random variable with $\mu=0$ and $\sigma^2=1$ is called a **standard normal random variable** and is denoted as Z .
- If X is a normal random variable with $E(X) = \mu$ and $V(X) = \sigma^2$, the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a standard normal random variable.

Probability Distributions and Probability Density Functions

The cumulative distribution function of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \text{ for } -\infty < x < \infty.$$

The Central Limit Theorem

- In particular, if $x_1, \dots, x_n$ are random variables with mean μ and standard deviation σ , and if n is large, then:

$$\frac{1}{n}(x_1 + \dots + x_n)$$

is approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.

Statistics

Statistics refers to the mathematics and techniques with which we understand data.

Describing a Single Set of Data

- Measuring the Central Tendency
- Measuring the Dispersion of Data

Describing a Single Set of Data

Measuring the Central Tendency

- mean (or average): the sum of the data divided by its count
- median: the middle number; the value less than which 50% of the data lies.
 - ordering all data points
 - picking out the one in the middle (or if there are two middle numbers, taking the mean of those two numbers).
- mode: most-common values

1, 3, 3, **6**, 7, 8, 9

Median = **6**

1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$
= **4.5**

Describing a Single Set of Data

- Positively skewed: the mode occurs at a value that is smaller than the median
- Negatively skewed: the mode occurs at a value greater than the median

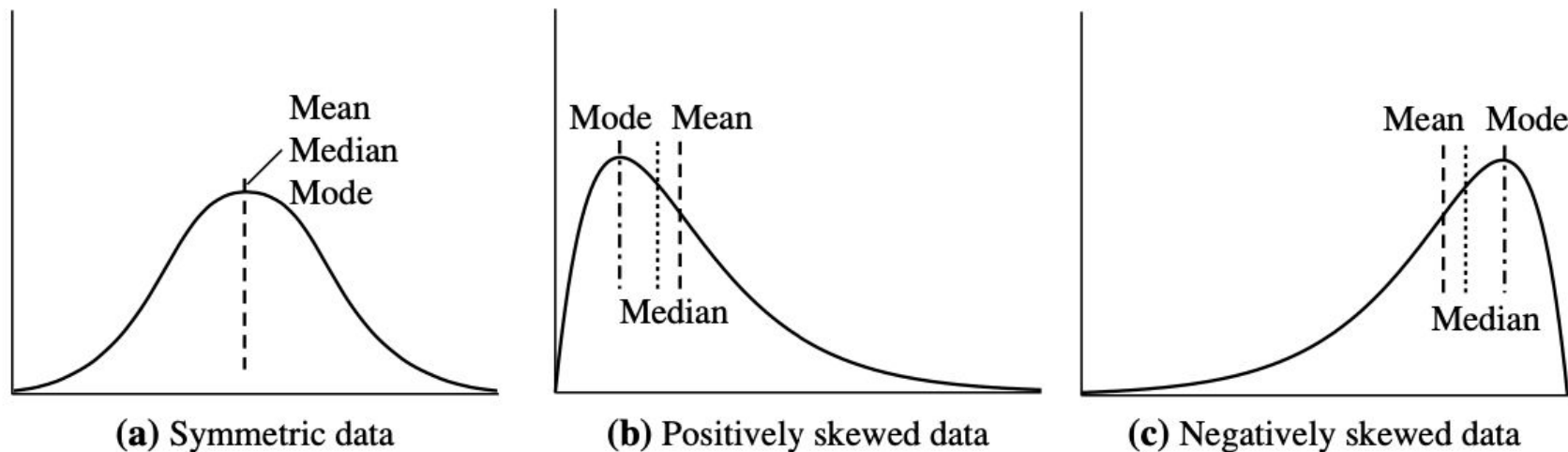


Figure 2.1 Mean, median, and mode of symmetric versus positively and negatively skewed data.

Describing a Single Set of Data

Measuring the Dispersion of Data: measure how spread out our data is.

- Data range: the difference between the largest ($max()$) and smallest values ($min()$)
- Quantile: split the data distribution into equal-size consecutive sets, these data points are called quantiles
 - The 2-quantile is **the data point** dividing the lower and upper halves of the data distribution. It corresponds to the median.
 - The 4-quantiles (quartiles) are **the three data points** that split the data distribution into four equal parts; each part represents one-fourth of the data distribution.
 - The 100-quantiles (percentiles) are **the 99 data points** that divide the data distribution into 100 equal-sized consecutive sets.

Describing a Single Set of Data

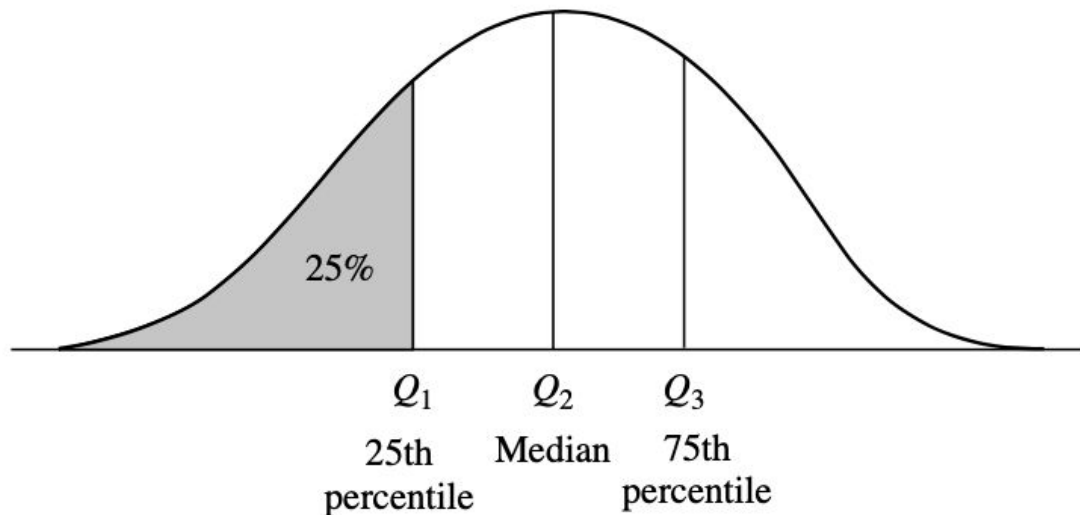


Figure 2.2 A plot of the data distribution for some attribute X . The quantiles plotted are quartiles. The three quartiles divide the distribution into four equal-size consecutive subsets. The second quartile corresponds to the median.

Describing a Single Set of Data

- Variance (σ^2): measure how far each number in the set is from the mean (average) of the dataset
- The standard deviation, σ , of the observations is the square root of the variance.

Statistical Hypothesis Testing

- We'll want to test whether a certain hypothesis is likely to be true.
- A null hypothesis H_0 that represents some default position (“no difference” or “no effect.”), and some alternative hypothesis H_1 that we'd like to compare it with.
- We use sample data and statistical tests to compute the *p-value* (the probability with assuming H_0 is true).
- If *p-values* is smaller than the 5% significance, we would reject the null. Otherwise, we accept the null.